

Logistic Regression with Pytorch Analysis:

1. How does the setup differ from the model you learned "by hand" (Logistic regression homework) in terms of initialization, number of parameters, activation?

The "by hand" implementation required fewer parameters for the buzzer initialization (the number of features, regularization parameter, the learning rate), whereas the optimized version with pytorch required the number of features, regularization parameter, learning rate, beta parameters for Adam, and epsilon parameter for Adam. As a result, the update process for the optimized model required more computational overhead than for the "by hand" version, as more parameters and calculations were needed. Moreover, the "by hand" activation took a single float value as an input (score), whereas for the optimized implementation, the input to torch.sigmoid was a tensor array.

2. Look at the top features again. Do they look consistent with your results for the last homework? Hint: run `pytorch_custom_adam_buzzer.py` and `toylogistic_buzzer.py` to compare the magnitude of the top features

The top features for the pytorch logistic regression were:

INFO:root:Feat	Gpr_confidence	11: +7.76091
INFO:root:Feat	Category_category:Geography	2: +2.07998
INFO:root:Feat	Frequency_guess	10: +1.58366
INFO:root:Feat	Length_char	12: +1.32969
INFO:root:Feat	Length_ftp	13: +0.43107
INFO:root:Feat	Length_guess	14: -1.02101
INFO:root:Feat	Category_category:Literature	4: -1.14159
INFO:root:Feat	Category_category:Fine Arts	1: -1.84893
INFO:root:Feat	Category_category:Mythology	5: -2.10369
INFO:root:Feat	Category_category:Religion	7: -2.68220

The top features for the manual logistic regression were:

INFO:root:Feat	Gpr_confidence	11: +6.18014
INFO:root:Feat	BIAS_CONSTANT	0: +1.81244
INFO:root:Feat	Category_category:Geography	2: +1.59577
INFO:root:Feat	Frequency_guess	10: +1.59292
INFO:root:Feat	Length_char	12: +1.01112
INFO:root:Feat	Length_guess	14: -0.58774
INFO:root:Feat	Category_category:Religion	7: -0.63696
INFO:root:Feat	Category_category:Literature	4: -0.83833
INFO:root:Feat	Category_category:Fine Arts	1: -0.91764
INFO:root:Feat	Category_category:Mythology	5: -1.20754

The features themselves were almost identical, with the exception of `Length_ftp` being present in the pytorch implementation and not in the manual implementation, as well as the `BIAS_CONSTANT` being listed as a feature for the manual implementation. Moreover, though the exact numbers differ, the sign associated with each feature (positive/negative) remained consistent across both implementations.